

BUSINESS REQUIREMENT DOCUMENT

Expense Tracker With Smart Insider

AI-Powered Personal Finance Management Platform

1. Introduction

The Smart Expense Tracker Smart Insider is a personal finance application built to transform how individuals manage their budgets. Standard financial tools only record transactions reactively. This platform bridges that gap by using Machine Learning to deliver proactive spending forecasts and cluster-based behavioural analysis.

2. Project Overview & Tech Stack

Component	Technology
Frontend	React
Backend	Python (Fast API)
Database	PostgreSQL
Authentication	JWT (JSON Web Tokens)
API Architecture	RESTful APIs / OpenAPI (Swagger)
Machine Learning	Python (Scikit-Learn)

3. Business Problem Statement

Many individuals struggle with tracking personal finances, managing recurring expenses, and controlling impulse spending. Conventional budgeting solutions require manual calculation to project future expenses, leading to delayed financial decisions, unexpected budget overruns, and a lack of actionable insight into spending habits.

4. Business Objectives

- **Automated Predictive Forecasting:** Leverage Scikit-learn regression models to forecast short-term and long-term spending trends.
- **Behavioural Clustering:** Group transaction patterns into spending clusters (e.g., weekend shopping, recurring bill commitments) using K-Means clustering.
- **Secure Financial Data Management:** Ensure user data protection through password hashing (Bcrypt), JWT token authorization, and PostgreSQL data isolation.
- **High-Performance Architecture:** Maintain good response times for financial CRUD operations and low-latency execution for ML inference.

5. Functional Requirements

5.1 User Management & Security

- User registration, secure login, and session persistence via JWT access and refresh tokens.
- User profile management with currency and budget preference settings.

5.2 Financial Record Management

- Complete CRUD operations (Create, Read, Update, Delete) for income and expense transactions.
- Custom and pre-defined category assignments (e.g., Utilities, Food, Transportation, Subscriptions).

5.3 Smart Analytics & ML Engine

- Expense forecasting over 7-day and 30-day time horizons using Scikit-learn.
- Spending habit clustering to highlight user behavioral archetypes.
- Automated anomaly detection for transactions deviating significantly from historical baselines.

5.4 Interactive Dashboard

- Dynamic visual charts built with React displaying net balance, category breakdowns, cash flow, and predicted trajectories.
- Data export functionality (CSV/PDF) for offline record keeping.

6. Non-Functional Requirements

- **Performance:** REST API; ML analytical inference.
- **Scalability:** Stateless Fast API backend architecture designed for horizontal scaling.
- **Security:** Mandatory HTTPS, short-lived JWTs, salted password storage, and SQL injection prevention via SQLAlchemy ORM.
- **Reliability:** 99.5% service availability with automated fallback strategies when transaction counts are insufficient for ML inference.

7. User Roles & Responsibilities

Role	Responsibilities
End User	Logs daily income/expense records, creates custom categories, views financial analytics, and monitors ML spending forecasts.
System Administrator	Oversees the expenses of all the member, manages user access controls, monitors API usage, and ensures database backup compliance.
Data Science Lead	Evaluates Scikit-learn model accuracy, updates feature engineering pipelines, and fine-tunes clustering parameters.

8. Project Workflow

1. Client Interaction: User signs in via the React frontend and sends a request containing credentials to the Fast API authentication endpoint.
2. Authentication: Fast API verifies credentials against PostgreSQL and returns signed JWT access tokens.
3. Data Logging: Users add income and expense entries, which are stored securely in PostgreSQL using SQLAlchemy.
4. ML Inference: Historical records are processed through Scikit-learn pipelines to calculate trend predictions and cluster spending behaviour.
5. Dashboard Rendering: Visual updates, predicted charts, and expenditure alerts are streamed back to the React UI.

9. Future Enhancements

- **Bank Aggregation APIs:** Automated transaction syncing via secure Open Banking interfaces.
- **Automated Receipt Parsing:** Optical Character Recognition (OCR) to automatically log expenses from receipt images.
- **Multi-Currency & FX Updates:** Real-time exchange rate updates for international multi-currency transaction conversion.
- **Push Notifications:** Automated alerts for budget limit thresholds and anomalous transaction flags.

10. Conclusion

The Smart Expense Tracker & Predictive Financial Insider addresses key shortcomings of traditional personal finance applications by combining a lightweight full-stack structure with intelligent machine learning analytics. Utilizing React, Fast API, PostgreSQL, and Scikit-learn ensures a reliable, scalable, and predictive environment for modern budget management.